Question 4 — Dining Philosophers: Deadlock-Avoidance Write-Up

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The deadlock this problem invites

Each fork is a binary semaphore shared by two neighbouring philosophers. If every philosopher picks up their *left* fork first, then their *right* fork second, and all five happen to grab their left fork at roughly the same time, every philosopher now holds exactly one fork and is blocked waiting on their neighbour's fork — a circular wait with no philosopher able to make progress. That is the classic Dining Philosophers deadlock.

Strategy used: resource ordering

Philosopher `i` (0..4) has neighbouring forks `left = i` and `right = (i+1) % 5`. Instead of always taking left-then-right, each philosopher computes `first = min(left, right)` and `second = max(left, right)`, and always acquires the *lower-numbered* fork first, regardless of which physical side it sits on:

```
int left = id, right = (id + 1) % 5;
int first  = left < right ? left  : right;
int second = left < right ? right : left;
sem_wait(first);
sem_wait(second);
```

Why this prevents deadlock

A circular wait requires every fork in some cycle to be held by one philosopher while that philosopher waits on the next fork in the cycle. With a strict global ordering on fork acquisition, fork 4 (the highest-numbered fork) is never acquired before some lower-numbered fork by any philosopher who needs it — so at least one philosopher in any would-be cycle is always trying to acquire a *lower*-numbered fork first, which some other philosopher already holds only as their *second* fork, not while waiting on a third. This breaks the necessary-condition chain for deadlock (Coffman's circular-wait condition): the wait-for graph among forks can no longer contain a cycle, because every edge in it respects the same total order.

Result

Run with `CPUS=1`, all 5 philosophers complete all 5 THINKING → HUNGRY → EATING → THINKING cycles (25 EATING events total), the process never hangs, and no philosopher is starved for the duration of the run — confirmed by the full transcript in `output_logs/dining_full.txt` and the screenshot in `screenshots/q4_dining.png`, ending with `All philosophers finished 5 cycles, no deadlock`.